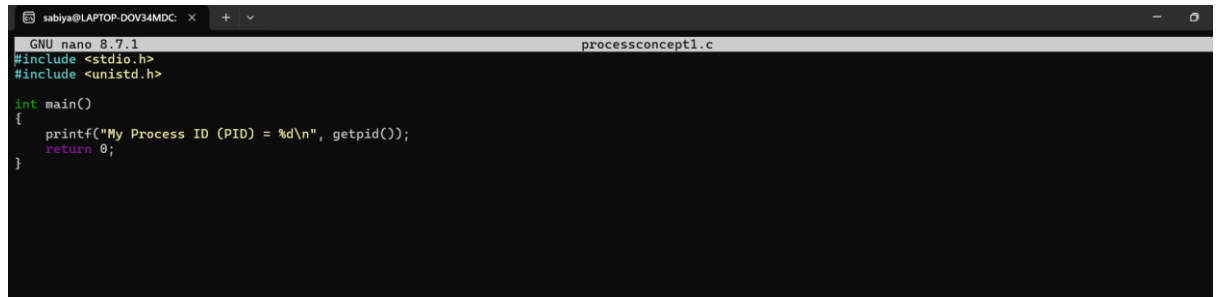


OSP-SP Process Concepts Problems (CODES)

2520030502

Sk.sabiya thabassum

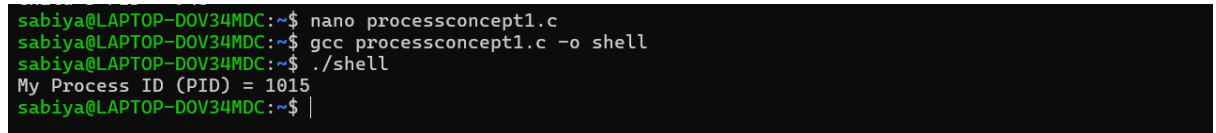
1. Process id



```
GNU nano 8.7.1 processconcept1.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("My Process ID (PID) = %d\n", getpid());
    return 0;
}
```

Output Screenshot:



```
sabiya@LAPTOP-DOV34MDC:~$ nano processconcept1.c
sabiya@LAPTOP-DOV34MDC:~$ gcc processconcept1.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
My Process ID (PID) = 1015
sabiya@LAPTOP-DOV34MDC:~$ |
```

2.Question



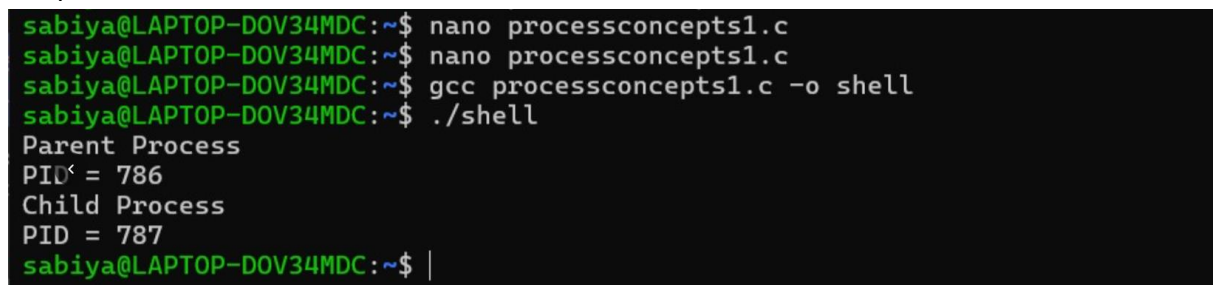
```
GNU nano 8.7.1 processconcepts1.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if(pid == 0)
    {
        printf("Child Process\n");
        printf("PID = %d\n", getpid());
    }
    else
    {
        printf("Parent Process\n");
        printf("PID = %d\n", getpid());
    }

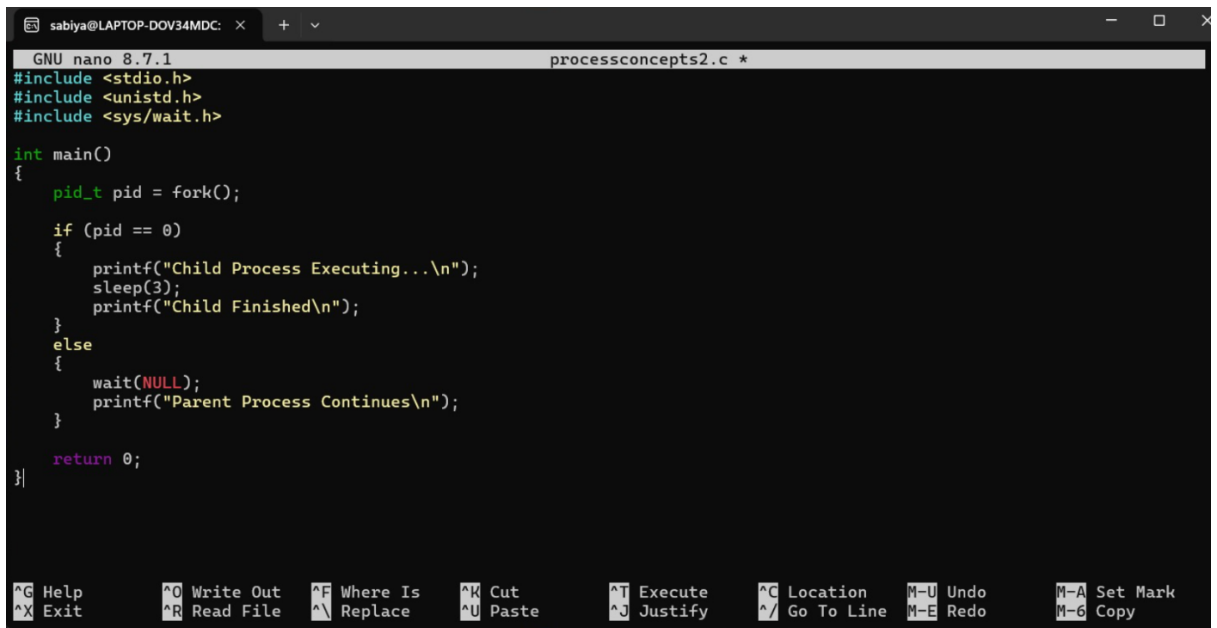
    return 0;
}
```

Output Screenshot:



```
sabiya@LAPTOP-DOV34MDC:~$ nano processconcepts1.c
sabiya@LAPTOP-DOV34MDC:~$ nano processconcepts1.c
sabiya@LAPTOP-DOV34MDC:~$ gcc processconcepts1.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Parent Process
PID = 786
Child Process
PID = 787
sabiya@LAPTOP-DOV34MDC:~$ |
```

3.



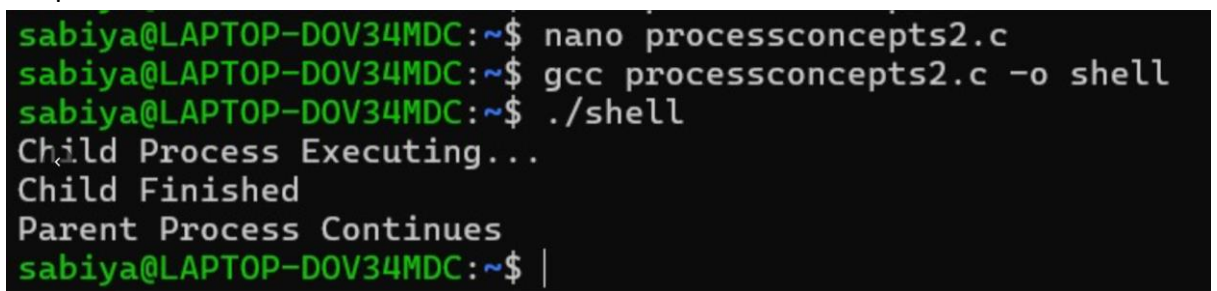
```
GNU nano 8.7.1 processconcepts2.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child Process Executing...\n");
        sleep(3);
        printf("Child Finished\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Process Continues\n");
    }

    return 0;
}
```

Output Screenshot:



```
sabiya@LAPTOP-DOV34MDC:~$ nano processconcepts2.c
sabiya@LAPTOP-DOV34MDC:~$ gcc processconcepts2.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Child Process Executing...
Child Finished
Parent Process Continues
sabiya@LAPTOP-DOV34MDC:~$ |
```

4. Multiple child processes



```
GNU nano 8.7.1 processconcepts3.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    for(int i = 1; i <= 3; i++)
    {
        if(fork() == 0)
        {
            printf("Child %d PID = %d\n", i, getpid());
            return 0;
        }
    }

    return 0;
}
```

Output Screenshot:

```
sabiya@LAPTOP-DOV34MDC:~$ nano processconcepts3.c
sabiya@LAPTOP-DOV34MDC:~$ nano processconcepts3.c
sabiya@LAPTOP-DOV34MDC:~$ gcc processconcepts3.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Child 1 PID = 944
Child 2 PID = 945
Child 3 PID = 946
sabiya@LAPTOP-DOV34MDC:~$ |
```

5. Excute linux command using extp()



```
GNU nano 8.7.1 processconcepts4.c
#include <stdio.h>
#include <unistd.h>

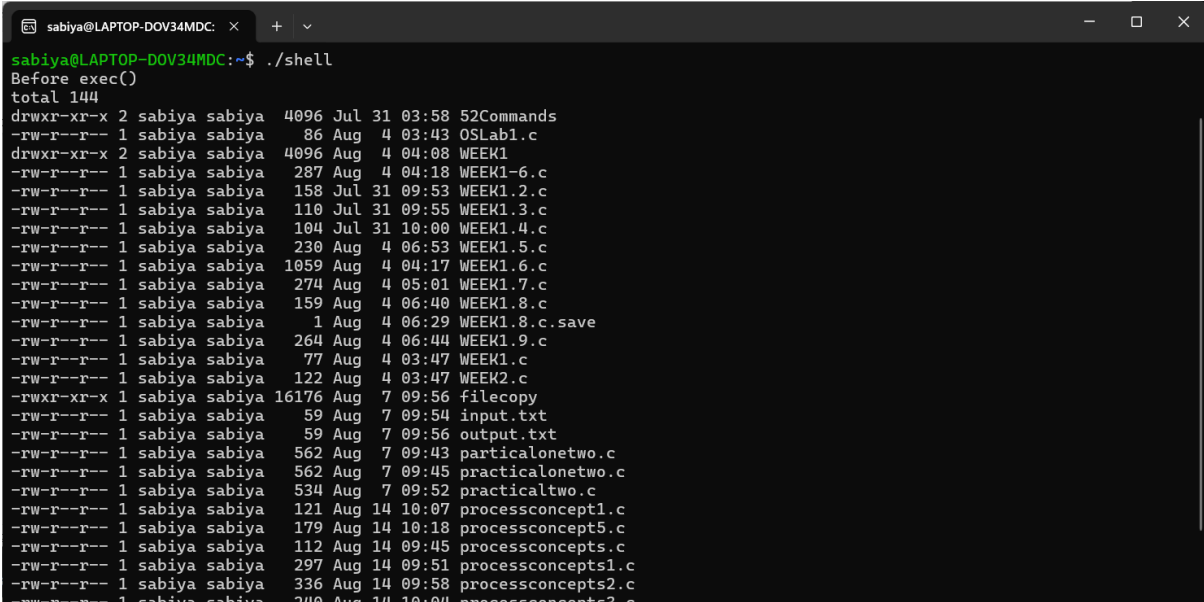
int main()
{
    printf("Before exec()\n");

    execlp("ls", "ls", "-l", NULL);

    printf("This will not execute.\n");

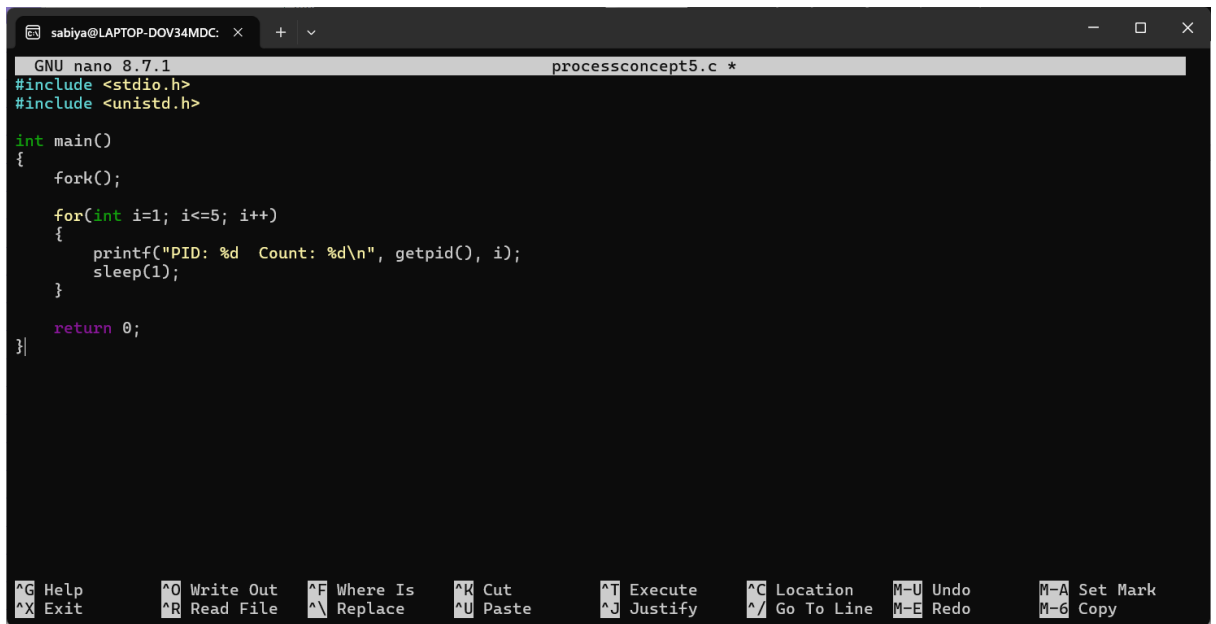
    return 0;
}
```

Output Screenshot:



```
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Before exec()
total 144
drwxr-xr-x 2 sabiya sabiya 4096 Jul 31 03:58 52Commands
-rw-r--r-- 1 sabiya sabiya 86 Aug 4 03:43 OSLab1.c
drwxr-xr-x 2 sabiya sabiya 4096 Aug 4 04:08 WEEK1
-rw-r--r-- 1 sabiya sabiya 287 Aug 4 04:18 WEEK1-6.c
-rw-r--r-- 1 sabiya sabiya 158 Jul 31 09:53 WEEK1.2.c
-rw-r--r-- 1 sabiya sabiya 110 Jul 31 09:55 WEEK1.3.c
-rw-r--r-- 1 sabiya sabiya 104 Jul 31 10:00 WEEK1.4.c
-rw-r--r-- 1 sabiya sabiya 230 Aug 4 06:53 WEEK1.5.c
-rw-r--r-- 1 sabiya sabiya 1059 Aug 4 04:17 WEEK1.6.c
-rw-r--r-- 1 sabiya sabiya 274 Aug 4 05:01 WEEK1.7.c
-rw-r--r-- 1 sabiya sabiya 159 Aug 4 06:40 WEEK1.8.c
-rw-r--r-- 1 sabiya sabiya 1 Aug 4 06:29 WEEK1.8.c.save
-rw-r--r-- 1 sabiya sabiya 264 Aug 4 06:44 WEEK1.9.c
-rw-r--r-- 1 sabiya sabiya 77 Aug 4 03:47 WEEK1.c
-rw-r--r-- 1 sabiya sabiya 122 Aug 4 03:47 WEEK2.c
-rwxr-xr-x 1 sabiya sabiya 16176 Aug 7 09:56 filecopy
-rw-r--r-- 1 sabiya sabiya 59 Aug 7 09:54 input.txt
-rw-r--r-- 1 sabiya sabiya 59 Aug 7 09:56 output.txt
-rw-r--r-- 1 sabiya sabiya 562 Aug 7 09:43 particalonnetwo.c
-rw-r--r-- 1 sabiya sabiya 562 Aug 7 09:45 practicalonnetwo.c
-rw-r--r-- 1 sabiya sabiya 534 Aug 7 09:52 practicaltwo.c
-rw-r--r-- 1 sabiya sabiya 121 Aug 14 10:07 processconcept1.c
-rw-r--r-- 1 sabiya sabiya 179 Aug 14 10:18 processconcept5.c
-rw-r--r-- 1 sabiya sabiya 112 Aug 14 09:45 processconcepts.c
-rw-r--r-- 1 sabiya sabiya 297 Aug 14 09:51 processconcepts1.c
-rw-r--r-- 1 sabiya sabiya 336 Aug 14 09:58 processconcepts2.c
-rw-r--r-- 1 sabiya sabiya 240 Aug 14 10:04 processconcepts3.c
```

6. Parent and Child Executing Concurrently



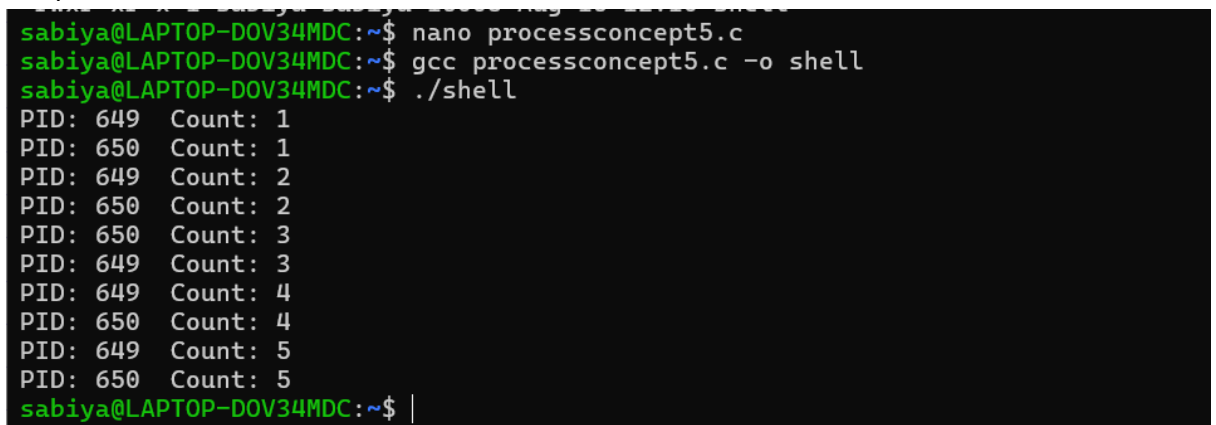
```
GNU nano 8.7.1 processconcept5.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();

    for(int i=1; i<=5; i++)
    {
        printf("PID: %d  Count: %d\n", getpid(), i);
        sleep(1);
    }

    return 0;
}
```

Output Screenshot:



```
sabiya@LAPTOP-DOV34MDC:~$ nano processconcept5.c
sabiya@LAPTOP-DOV34MDC:~$ gcc processconcept5.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
PID: 649  Count: 1
PID: 650  Count: 1
PID: 649  Count: 2
PID: 650  Count: 2
PID: 650  Count: 3
PID: 649  Count: 3
PID: 649  Count: 4
PID: 650  Count: 4
PID: 649  Count: 5
PID: 650  Count: 5
sabiya@LAPTOP-DOV34MDC:~$
```